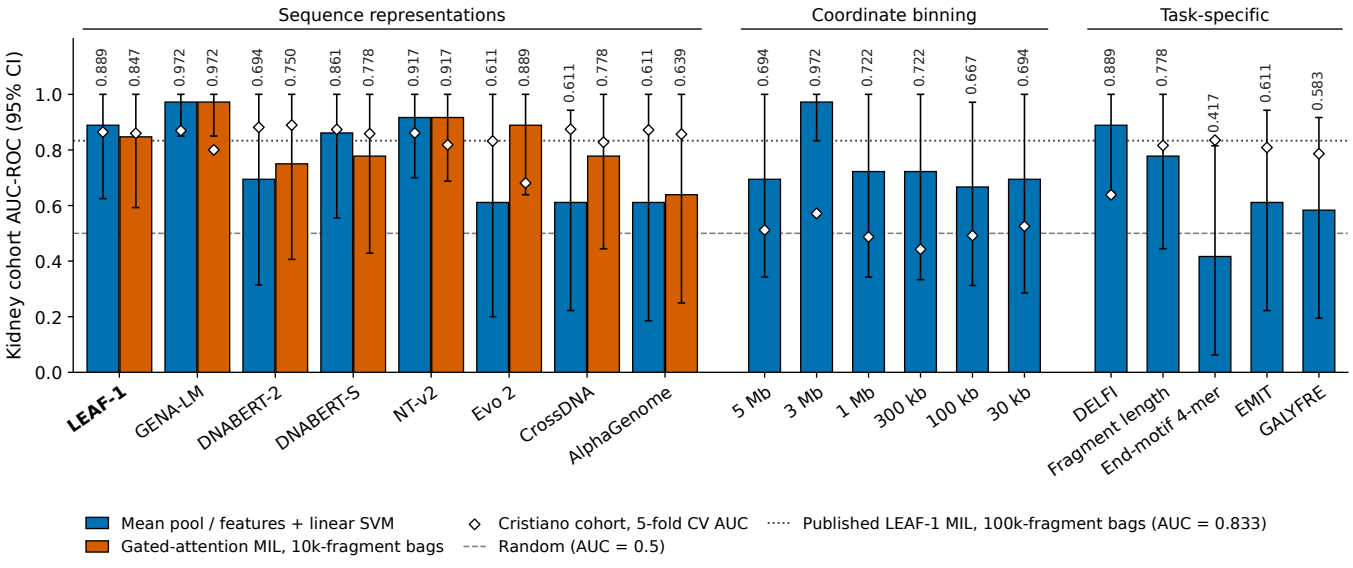
Independent kidney cohort, zero-shot, 10k fragments
Sequence representations
Coordinate binning
Task-specific
Kidney cohort AUC-ROC (95% CI)
0.0
0.2
0.4
0.6
0.8
1.0
0.889
0.847
0.972
0.972
0.694
0.750
0.861
0.778
0.917
0.917
0.611
0.889
0.611
0.778
0.611
0.639
0.694
0.972
0.722
0.722
0.667
0.694
0.889
0.778
0.417
0.611
0.583
LEAF-1
GENA-LM
DNABERT-2
DNABERT-S
NT-v2
Evo 2
CrossDNA
AlphaGenome
5 Mb
3 Mb
1 Mb
300 kb
100 kb
30 kb
DELFI
Fragment length
End-motif 4-mer
EMIT
GALYFRE
Mean pool / features + linear SVM
Gated-attention MIL, 10k-fragment bags
Cristiano cohort, 5-fold CV AUC
Random (AUC = 0.5)
Published LEAF-1 MIL, 100k-fragment bags (AUC = 0.833)